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1. A digital circuit that can store only one bit is a/an
(a) register (b) NOR gate
(c) flip-flop (d) XOR gate

**Computer – Flip-Flops – 2021**

2. How much input and output needed for demultiplexer?
(a) Many inputs one output
(b) One input many outputs
(c) One input one output
(d) None of the above

**Computer – Demultiplexer – 2021**

3. If A, B and C are the inputs of a full adder, then the sum is given by
(a) A and B And C (b) A OR B And C
(c) A XOR B XOR C (d) A OR B OR C

**Computer – Full Adder – 2021**

4. A DFlip-flop can be constructed from a/an ____flip-flop.
(a) S-R (b) J-K
(c) T (d) S-K

**Computer – Flip-Flop Conversion – 2021**

5. How many full adder and half adder are required to add 16-bit numbers?
(a) 8 half adders, 8 full adders
(b) 1 half adder, 15 full adders
(c) 16 half adders, no full adder
(d) 4 half adders, 12 full adders

**Computer – Adders – 2021**

6. Which flip-flops serves to be the fundamental building blocks of counters?
(a) S-R flip-flops (b) J-K flip-flops
(c) T flip-flops (d) D flip-flops

**Computer – Counters – 2021**

7. What is the difference between a shift-right register and a shift-left register?
(a) There is no difference
(b) The direction of the shift
(c) Propagation delay
(d) The clock input

**Computer – Shift Registers – 2021**

8. Which is not characteristic of a shift register?
(a) Serieal in/parallel in
(b)Serial in/ parallel out
(c) Parallel in / serial out
(d) Parallel in / Parallel out

**Computer – Shift Register Types – 2021**

9. In a processor, while executing an instruction ________
A. Programme Counter is used to hold the address of next instruction.
B. Instruction register holds the instruction for execution.
C. Memory Address Register is used to perform address translation.
D. Memory Data Register is used to perform data operation.
E. Clock generates control signals.
Choose the correct answer form the options given below.
(a) Only A and B are true
(b) Only C and D are true
(c) Only A, B and E are true
(d) Only B, C and D are true

**COMPUTER – CPU Registers – 2022**

**10.**
A. If $(12P)_3$-$(123)_p$, then value of P is infeasible.
B. The simplified sum of product form of the Boolean expression is $(P+\bar{Q}+\bar{R}).(P+\bar{Q}+R).(P+Q+\bar{R})\ is\ (P+\bar{Q}.\bar{R})$.
C. The minimum number of D flip-flops needed to design a mod (258) counter is 8.
Choose the correct answer from the options given below:
(a) A only (b) A and B only
(c) A and C only (d) C only

**COMPUTER** – Digital Electronics **– 2023**

11. Match List-I with List-II

| List-I | | List-II | |
|---|---|---|---|
| **A.** | Asynchronous | **I.** | A pulse that cause a logic device to be activated or change state. |
| **B.** | Trigger | **II.** | The operation is not executed in step with the clock. |
| **C** | J-K Flip-flop | **III.** | Flip Flop that atleast set, reset, toggle and hold modes of operation. |
| **D.** | D flip flop | **IV.** | Flip flop with atleast set and reset modes of operations. |

Choose the correct answer from the options given below:
(a) A-I, B-II, C-III, D-IV (b) A, II, B-I, C-III, D-IV
(c) A-III, B-IV, C-II, D-I (d) A-I, B-II, C-IV, D-III

Digital Electronics – Flip-Flops – 2024

12. Match List-I with List-II

| List-I | | List-II | |
|---|---|---|---|
| A. | Flash memory | I. | Oldest and Slowest |
| B. | PMOS | II. | Used in large scale integration (LSI) |
| C | NMOS | III. | Least power consumption. |
| D. | CMOS | IV. | Non volatile RAM which is powered continuously. |

Choose the correct answer from the options given below

(a) A-I, B-IV, C-III, D, II (b)A-I, B-IV, C-II, D-III
(c) A-IV, B-I, C-III, D-II (d) A-IV, B-I,C-II, D-III

Computer Organization – Memory Technologies – 2024

13. Consider the following statements
(A) RAM is a combinational circuit
(B) RAM is sequential circuit
(C) PLA is a combinational circuit
(D) PLA is a sequential circuit

(a) A and D only (b) B and C only
(c) B and D only (d) A and C only

Digital Logic – Combinational vs Sequential Circuits – 2024

14. A logic circuit, that can add two 1-bit numbers and produce output for sum and carry but cannot handle carry input, is called _______

(a) Half adder (b) Full adder
(c) Multiplexer (d) Encoder

Digital Electronics – Adders (Half Adder) – 2024

15. Minimum number of 2:1 Multiplexers required to design a 16:1 Multiplexer is?

(CUET-PG 2025) Multiplexar – 2025

(a)12 (b)14 (c)15 (d)16

16. Consider the following statements.
(a) Combinational logic circuits do not have memory, while sequential logic circuits have memory elements,
(b) Sequential logic circuits depend only on the current inputs, while combinational logic circuits depend on both current and past inputs.
(c) Flip-flops and latches are examples of combinational logic circuits.
(d) Multiplexer is an example of combinational logic circuit.

Choose the **correct** statement/statements from the options given below:

(CUET-PG 2025) Sequential Circuts – 2025

(a) (A) and (D) only
(b) (A), (B) and (D) only
(c) (A), (B), (C) and (D) only
(d) (B), (C) and (D) only

17. Arrange the following steps in the correct order to understand the functioning of a 3 to 8 line decoder.
(a) The encoder enable inputs is set to 1.
(b) The decoder activates one of its 8 output lines based on the input code.
(c) The input binary code is applied to the decoder.
(d) The decoder converts the 3- bit binary input into 8 possible outputs.

(CUET-PG 2025) Decodar – 2025

Choose the **correct** answer from the options given below:

(a)(A), (B), (C), (D) (b)(A), (D), (C), (B)
(c)(A), (C), (D), (B) (d)(A), (B), (D), (C)

44. Math List-I with List-II

| List-I | List-II |
|---|---|
| **J K Flip Flop inputs** | **Next State** |
| (a) 0 0 | (i). Toggle |
| (b) 0 1 | (ii). Set |
| (c) 1 0 | (iii). Reset |
| (d) 1 1 | (iv). No Change |

(CUET-PG 2025) Digital Electronics Flip Flop – 2025

Choose the correct answer from the options given below.

(a) (A) – (i), (B) – (ii), (C) – (iii), (D) – (iv)
(b) (A) – (iv), (B) – (iii), (C) – (ii), (D) – (i)
(c) (A) – (iv), (B) – (iii), (C) – (i), (D) – (ii)
(d) (A) – (i), (B) – (ii), (C) – (iv), (D) – (iii)

## ANSWER KEY

| 1. | 2. | 3. | 4. | 5. | 6. | 7. | 8. | 9. | 10. |
|---|---|---|---|---|---|---|---|---|---|
| c | b | c | a | b | c | b | a | c | b |
| 11. | 12. | 13. | 14. | 15. | 16. | 17. | 18. | | |
| b | d | b | a | c | a | c | b | | |